

Policy-Dependent Minimal Incompatibility of Synchronous-to-Inverter Replacement Portfolios: A Transverse Spectral Hypergraph and Its Network-Closure Anatomy

Jorge L. Mayorga Taborda

Abstract—Replacing synchronous generators (SGs) by grid-following inverters is planned unit by unit, yet small-signal stability is not additive: every subset of a portfolio can be stable while the portfolio is not. We formulate this as a minimal incompatibility hypergraph over binary replacement decisions and study it on a nonlinear phasor differential–algebraic model. Because frozen benchmark models lack primary frequency control, stability is posed exactly in transverse coordinates, as a quotient by the rotation and common-frequency-drift center subspace, without discarding small eigenvalues. We prove that the hypergraph changes only where a transverse eigenvalue crosses the imaginary axis, that it characterizes the portfolios that are safe in every implementation order, and that all non-additivity of the characteristic function is the principal-minor expansion of a network-closure operator between the replaced buses. A symmetry-deflated port extends the determinant test to zero-frequency crossings. Seven preregistered synthetic Monte Carlo suites and four suites on the IEEE 39-bus system test every statement: exact nonlinear symmetries, 80/80 time-domain verdicts, and a zero-frequency port holdout. One IEEE-39 rule failed as implemented and is reported with its post-hoc correction. On IEEE-39, the converter reactive policy changes which coalitions fail: 30, 36, and 16 distinct hypergraphs appear on three policy planes. A 2096.6-MW (4270.7-MVA) four-unit portfolio is unstable although every proper subset is stable, and each of ten frozen boundaries is closed only by the full-order network interaction of the changing coalition. With the source’s documented turbine governors the structure persists in a smaller region, while the four-unit coalition is stabilized.

Index Terms—Differential–algebraic equations, eigenvalue analysis, grid-following inverters, hypergraphs, low-inertia power systems, Monte Carlo validation, small-signal stability, synchronous generator retirement.

I. INTRODUCTION

GENERATION portfolios are decarbonized by replacing synchronous generators (SGs) with inverter-based resources, typically plant by plant. Small-signal stability of the resulting system is, however, a property of the *portfolio*. Low-inertia and converter-driven instabilities are now part of the stability taxonomy [1]–[3], and screening indices such as short-circuit ratios, single-port impedance criteria, or generalized short-circuit ratios [4]–[8] evaluate a configuration, not the combinatorial structure of a replacement plan. A planner needs a different answer: *which minimal groups of*

replacements are mutually incompatible, and how does that set depend on the converter control policy?

This paper studies that object. For a candidate set of m SGs and a converter policy θ , the minimal incompatibility hypergraph $\mathcal{H}(\theta)$ collects the inclusion-minimal replacement portfolios that are unstable. Its minimum edge size $\kappa(\theta)$ is the smallest incompatible coalition. Two technical obstacles appear immediately. First, frozen benchmark models (IEEE-39, Kundur, IEEE-68) have no rotor damping and no governors; their linearizations carry an exact double zero eigenvalue, so “stable” is undefined unless the neutral directions are handled exactly rather than by a magnitude cutoff. Second, a label per portfolio says nothing about *why* a coalition fails when its subsets do not.

Contributions.

- 1) A transverse stability notion (Theorem 1): the rotation and common-frequency drift span an exactly invariant Jordan center subspace; the quotient is exact for the linearization and for the nonlinear reduced field, so no eigenvalue is discarded by magnitude.
- 2) Structural properties of $\mathcal{H}(\theta)$ (Propositions 1 and 2, Theorem 2): antichain, boundary localization at imaginary-axis crossings, and the equivalence between “hyperedge-free” and “safe in every implementation order”; a monotone tractable class (Theorem 3); and the credited NP-hardness of minimum-cardinality destabilization (Proposition 3).
- 3) A network-closure factorization of the characteristic function (Theorem 4): on an exact descriptor realization that is affine in the replacement indicators, all non-additivity is a principal-minor expansion of an interaction operator Q generated by the network between the replaced buses; boundaries are zeros of $\det(I + Q_{SS})$, and their motion follows an eigenvalue-derivative formula (Proposition 4). A symmetry-deflated port (Theorem 5) extends the determinant test to crossings through $s = 0$.
- 4) Preregistered Monte Carlo validation: seven synthetic suites built to satisfy each theorem’s hypotheses, and four suites over random policies and portfolios of the IEEE 39-bus system, including nonlinear time-domain simulation (TDS).
- 5) IEEE-39 evidence: policy-dependent witness structure, the anatomy of ten frozen boundaries, robustness to

J. L. Mayorga Taborda is with the Department of Electrical and Electronic Engineering, Universidad de los Andes, Bogotá, Colombia (e-mail: jl.mayorga@uniandes.edu.co).

documented governors, and planning consequences, with every result stated under its model scope.

What is not claimed. The eigenvalue test (generalized Nyquist [9], [10]), Schur complements and Sylvester's determinant identity [11], Möbius inversion [12], clutters and minimal non-faces [13], minimal failure sets in $N-k$ analysis [14], and positive-systems monotonicity [15] are classical. The complexity statement is a corollary of the sparse-PCA reduction [16]. The contribution is the object $\mathcal{H}(\theta)$, its exact transverse definition, its network-closure anatomy, and the measured policy dependence.

II. MODEL AND PROBLEM FORMULATION

A. Phasor DAE

The system is the index-one DAE

$$\dot{x} = f(x, z; \delta, \theta), \quad 0 = g(x, z; \delta, \theta), \quad (1)$$

where $x \in \mathbb{R}^n$ collects device states, $z \in \mathbb{R}^{2N}$ the rectangular bus voltages, and g is Kirchhoff's current law with the bus admittance matrix, device injections, and constant-power loads. Index one means that g_z is nonsingular along the trajectories of interest. At an equilibrium the reduced Jacobian is $A = f_x - f_z g_z^{-1} g_x$.

Synchronous generators. Each SG uses a two-axis model on its own base [17]. With (v_d, v_q) the terminal voltage in the rotor frame and (i_d, i_q) obtained from the stator equations with transient reactances x'_d, x'_q ,

$$\dot{\delta} = \omega_B(\omega - 1), \quad M\dot{\omega} = P_m - P_e - D(\omega - 1), \quad (2)$$

$$T'_{d0}\dot{e}'_q = e_{fd} - e'_q - (x_d - x'_d)i_d,$$

$$T'_{q0}\dot{e}'_d = -e'_d + (x_q - x'_q)i_q,$$

with air-gap power $P_e = v_d i_d + v_q i_q + r_a(i_d^2 + i_q^2)$. The excitation is a first-order AVR $T_E \dot{e}_{fd} = K_A(v_{\text{ref}} + v_s - |V|) - e_{fd}$, and the stabilizer is the power-input washout-and-lag $v_s = K_S x_6, T_4 \dot{x}_6 = T_5(P_e - x_5)/T_6 - x_6, T_6 \dot{x}_5 = P_e - x_5$. As in the source data, $D = 0$ and P_m is constant (no governor).

Grid-following inverters. Each GFL has a synchronous-reference-frame PLL, filtered power measurements, PI outer loops, and PI inner current loops behind a filter reactance x_f :

$$\dot{\vartheta} = k_{p,\text{pll}} v_q + x_{\text{pll}}, \quad \dot{x}_{\text{pll}} = k_{i,\text{pll}} v_q, \quad (3)$$

$$\tau \dot{p}_f = P - p_f, \quad \tau \dot{q}_f = Q - q_f,$$

$$i_d^{\text{ref}} = k_p(p_{\text{ref}} - p_f) + x_p, \quad i_q^{\text{ref}} = -k_q(q_{\text{cmd}} - q_f) - x_q,$$

$$\dot{i}_{dq} = \frac{\omega_B}{x_f} (k_i(i_{dq}^{\text{ref}} - i_{dq}) + x_{i,dq} - r_f i_{dq}),$$

with PI states $\dot{x}_p = k_{ip}(p_{\text{ref}} - p_f)$, $\dot{x}_q = k_{iq}(q_{\text{cmd}} - q_f)$, and $\dot{x}_{i,dq} = k_{ii}(i_{dq}^{\text{ref}} - i_{dq})$. The injection is $\gamma(i_d + j i_q)e^{j\vartheta}$, with γ the rating weight. The reactive command comes from a leaky Q/V regulator

$$q_{\text{cmd}} = k_{p,v} g e + x_v, \quad e = v_{\text{ref}} - |V|, \quad (4)$$

$$\dot{x}_v = k_{i,v} g e - w(x_v - q_{\text{ref}}),$$

with leak $w = 0.05$ rad/s; $g = 0$ is exact fixed- Q operation. The parameters are listed in Table I. Converter limits, DC links, and modulation delay are not modeled, so nonlinear results are bounded by declared model-scope guards (Section VII-F).

TABLE I
DEVICE AND BENCHMARK DATA

Item	Value (per unit on the device rating)
SG model	two-axis, power-form swing, $D = 0$, constant P_m
AVR	$K_A = 10.1$, $T_E = 0.25$ s (harmonized IEEEEX1)
PSS	power input: $K_S = -2$, $T_5 = 1$, $T_6 = 4.2$, $T_4 = 0.75$ s
PLL	$k_{p,\text{pll}} = 53$, $k_{i,\text{pll}} = 1400$
Outer loops	$\tau = 0.03$ s; $k_p/k_{ip} = k_q/k_{iq} = 0.2/8$
Inner loops	$k_i/k_{ii} = 0.25/6$; $x_f = 0.15$, $r_f = 0.01$
Q/V regulator	$k_{p,v}/k_{i,v} = 2/20$; leak $w = 0.05$ rad/s; $g \in [0, 1]$
Loads	constant power
Network	IEEE 39-bus, 46 branches, 10 SGs (bus 39: interconnection)
Core	SGs 30, 33, 35, 37 (2096.6 MW; 4270.7 MVA)
Census	SGs 30–38 ($2^9 = 512$ portfolios)

TABLE II
NOTATION

Symbol	Meaning
V, m	candidate SGs, $ V = m$
S, δ_S	replacement portfolio and its indicator
$\theta = (g, k, t, h)$	policy: Q/V gain, AVR gain and time scales, heterogeneity
R_x, w	rotation generator and uniform frequency shift
$C, A_\perp$	center subspace and transverse operator
$\alpha_\perp(S, \theta)$	transverse spectral abscissa
$\mathcal{H}(\theta), \kappa(\theta)$	minimal incompatibility hypergraph, minimum edge size
A, B, C	final stable, safe one-at-a-time path, safe in every order
$T_S(s)$	port (network variables kept, device states eliminated)
$\Delta Y_i, K$	local replacement increment, base network closure
$M = DK, Q$	port matrix and interaction operator (6)
μ_T	Möbius (interaction) coefficient of order $ T $
r_S, R_k	tolerated severity of S , minimum over $ S \leq k$
ρ_{scope}	largest severity decidable inside the model's guards

B. Replacement portfolios and policy

Let $V = \{1, \dots, m\}$ be the candidate SGs. A portfolio $S \subseteq V$ has indicator $\delta_S \in \{0, 1\}^m$. Replacing unit i installs a GFL with the same active and reactive dispatch (*matched dispatch*) and rating equal to the SG rating, so the AC power flow is identical for every portfolio. The policy $\theta = (g, k, t, h)$ collects the Q/V gain g , a common scale k of every surviving SG's AVR gain, a scale t of every AVR time constant, and a heterogeneity exponent h ($T_{E,i} \mapsto G(T_{E,i}/G)^h$, with G the geometric mean). Active dispatch (MW) and converter rating (MVA) are always reported separately.

III. TRANSVERSE STABILITY

Assumption 1. (i) Every device angle state (rotor angle, PLL angle) enters (1) only through $V e^{-j\cdot}$, the other device states are written in their own rotating frames, and loads and network are rotation-covariant. (ii) Every SG has $D = 0$ and constant P_m , and the network is frequency-independent.

Let R_x equal 1 on every rotor and PLL angle and 0 elsewhere, and let w be the uniform frequency shift (speed +1, PLL integrator $+\omega_B$).

Theorem 1 (Exact transverse quotient). *Under Assumption 1(i), $AR_x = 0$. Under Assumption 1(ii), $Aw = \omega_B R_x$, so $C = \text{span}\{R_x, w\}$ is A -invariant with $A[R_x w] =$*

$[R_x \ w]J$, $J = \begin{bmatrix} 0 & \omega_B \\ 0 & 0 \end{bmatrix}$ (a Jordan chain). For any orthonormal Z with range $Z = C^\perp$ and $A_\perp = Z^\top AZ$,

$$\det(sI - A) = s^2 \det(sI - A_\perp), \quad (5)$$

and $\sigma(A_\perp)$ does not depend on Z . Moreover, the reduced field $r(x) = f(x, \psi(x))$, with ψ the local algebraic solution, satisfies $r(x + \phi R_x) = r(x)$ and $r(x + cw) = r(x) + c\omega_B R_x$ for all ϕ, c . Hence $\dot{y} = Z^\top r(x^* + Zy)$ is an exact reduced system whose linearization is $A_\perp$. If some SG has $D \neq 0$ or a governor, only $C = \text{span}\{R_x\}$ remains.

Proof. Rotation covariance gives $f(x + \phi R_x, e^{j\phi} z) = f(x, z)$ and a covariant g . Differentiating at $\phi = 0$ gives $f_x R_x + f_z R_z = 0$ and $g_x R_x + g_z R_z = 0$ with $R_z = jV$, so $AR_x = 0$. Under (ii), $x_0 + cw + c\omega_B t R_x$ with $z(t) = e^{j\omega_B t} z_0$ solves (1) exactly (electrical power is rotation invariant and the swing equation has $\dot{\omega} = 0$ at speed $1 + c$). Differentiating in c gives $Aw = \omega_B R_x$. In the basis $[U \ Z]$ with U an orthonormal basis of C , invariance gives $Z^\top AU = 0$, so A is block upper triangular with diagonal blocks $U^\top AU$ (spectrum $\{0, 0\}$) and $A_\perp$; this yields (5). A different complement ZO gives $O^\top A_\perp O$. The nonlinear identities follow from the same invariances applied to $f(x, \psi(x))$ with ψ rotating covariantly and unaffected by the uniform speed shift. Hence $Z^\top r(x + v) = Z^\top r(x)$ for $v \in C$. $\square$

Definition 1. A portfolio is *transversely stable* at θ if $\alpha_\perp(S, \theta) = \max \text{Re } \sigma(A_\perp(S, \theta)) < 0$.

Remark 1. The rotation is a gauge. The drift is a physical consequence of absent primary frequency control, and transverse stability is stability modulo absolute frequency restoration. A rule of the form “ignore $|\lambda| \leq \varepsilon$ ” coincides with Definition 1 only when $A_\perp$ has no physical eigenvalue in that disc. Suite S1 (Section VI) shows that it miscounts every draw with a planted *positive* physical mode of size 10^{-5} – 5×10^{-4} (235 of 235) and none of the others.

A. Numerical classification of transverse labels

Every label in this paper is produced by the procedure below. It has three outcomes and no magnitude cutoff.

- 1) Solve the AC equilibrium (Newton, residual $< 10^{-12}$) and form the DAE Jacobians by central differences at two step scales, h and $2h$. Their difference gives the Jacobian error E .
- 2) Build R_x and w from the device frames, verify $\|AR_x\|$ and $\|Aw - \omega_B R_x\|$, delete identically zero rows (switched-off services), and form $A_\perp = Z^\top AZ$ and $E_\perp = Z^\top EZ$.
- 3) If $\min |\text{Re } \sigma(A_\perp)| > 2 \times 10^{-3} \text{ s}^{-1}$, the label is the sign of $\alpha_\perp$.
- 4) Otherwise, compute the distance to the imaginary axis, $d = \min_\omega \sigma_{\min}(A_\perp - j\omega I)$, after balancing. The label is exact if $d > 10\varepsilon$ with $\varepsilon = \|E_\perp\| + \text{backward error}$. If not, the point is BOUNDARY-OR-UNRESOLVED and is reported, never assigned.

The criterion in step 4 is numerical, not an interval certificate. Because $\mathcal{H}(\theta)$ is a function of 2^m labels, a map with one

unresolved subset is flagged as unresolved. The frozen IEEE-39 maps carry 1.5% such points (Section VII).

IV. MINIMAL INCOMPATIBILITY HYPERGRAPH

Definition 2. For fixed θ with a transversely stable base, let $\mathcal{U}(\theta) = \{S \subseteq V : \alpha_\perp(S, \theta) \geq 0\}$. The *minimal incompatibility hypergraph* $\mathcal{H}(\theta)$ is the family of inclusion-minimal elements of $\mathcal{U}(\theta)$, and $\kappa(\theta) = \min_{e \in \mathcal{H}} |e|$ ($\kappa = \infty$ if $\mathcal{H} = \emptyset$).

Proposition 1 (Structure). $\mathcal{H}(\theta)$ is an antichain. A portfolio containing no hyperedge is transversely stable. The converse is false: $\mathcal{U}$ need not be upward closed.

Proof. Minimal elements of a family form an antichain. If S is unstable, a minimal unstable subset of S is a hyperedge contained in S . Suite S2 and the IEEE-39 census give stable supersets of unstable sets. $\square$

For a target T , consider three planning predicates. **A:** T is transversely stable. **B:** some one-at-a-time chain $\emptyset = S_0 \subset \dots \subset S_{|T|} = T$ has every S_j stable. **C:** every $R \subseteq T$ is stable, so every implementation order passes only through stable stationary states.

Proposition 2 (Planning hierarchy). $C \Rightarrow B \Rightarrow A$, and neither converse holds in general. With a stable base, T satisfies C if and only if T contains no hyperedge of $\mathcal{H}(\theta)$.

Proof. C gives any chain; B ends at T . T violates C iff some $R \subseteq T$ is unstable, iff (Proposition 1) some hyperedge lies in T . Suite S2 exhibits both converse failures. $\square$

Remark 2 (Examples). For A without B, take $m = 2$ and diagonal vertices $A_\emptyset = -I$, $A_{\{1\}} = \text{diag}(2, -3)$, $A_{\{2\}} = \text{diag}(-3, 2)$, and $A_{\{1,2\}} = -2I$. The target $\{1, 2\}$ is stable, but both one-at-a-time chains pass through an unstable singleton. For B without C, it suffices that one proper subset is unstable while another chain avoids it. On the IEEE-39 census, each order-dependent target has exactly one unstable subset (Section VII-G).

Theorem 2 (Boundary localization). Let $\theta(\tau)$, $\tau \in [0, 1]$, be continuous with g_z nonsingular along the path for every portfolio, and let Assumption 1 hold. Then $\mathcal{H}(\theta(\tau))$ is constant on every interval on which no $A_\perp(S, \theta(\tau))$ has an eigenvalue on the imaginary axis. Every change of $\mathcal{H}$ is bracketed by such a crossing, at finite frequency or at $s = 0$.

Proof. $A_\perp(S, \theta(\tau))$ is continuous in τ because g_z^{-1} is. Its eigenvalues are continuous, so the sign of $\alpha_\perp(S, \cdot)$ can change only through $\text{Re } \lambda = 0$. $\mathcal{H}$ is a function of the 2^m labels. $\square$

Theorem 3 (Monotone class). If there is an invertible T such that every $B(S) = T^{-1}A(S)T$ is Metzler and $S \subseteq S'$ implies $B(S) \leq B(S')$ entrywise, then $\alpha_\perp$ is monotone on the lattice. Stability is then hereditary, $\mathcal{H}$ consists of the minimal non-faces of the stable family, and $A = B = C$.

Proof. For Metzler B and $c \geq \max |B_{ii}|$, $\alpha(B) = \rho(B + cI) - c$; the spectral radius is monotone on nonnegative matrices [15], and α is similarity invariant. $\square$

A single robust pair $S \subset S'$ with $\alpha_\perp(S) > 0 > \alpha_\perp(S')$ excludes every such realization. IEEE-39 contains three such pairs (Section VII-G).

Proposition 3 (Complexity, credited). *Deciding whether some $|S| \leq k$ has $\alpha(A(\delta_S)) \geq 0$ is NP-complete, already for $A_G(\delta) = D_\delta W D_\delta - (k-1)I$ with W a graph adjacency matrix and $D_\delta = \text{diag}(\delta)$.*

Proof. $\alpha(A_G(\delta_S)) \geq 0$ with $|S| \leq k$ iff $\lambda_{\max}(W[S]) \geq k-1$, iff $G[S] = K_k$ [18]. Membership follows from exact $LDL^\top$. This is the clique reduction behind the NP-hardness of sparse PCA [16]; robust stability of interval and affine families is NP-hard as well [19], [20]. Structured radii under continuous perturbations are treated in [21]. $\square$

V. NETWORK-CLOSURE FACTORIZATION AND PORTS

A. Descriptor realization affine in the indicators

Under matched dispatch, every candidate bus can carry both devices: the SG with weight $(1 - \delta_i)$ and the GFL with weight δ_i . An absent device is replaced by a stable filler $\dot{x} = -(x - x_0)$ of the same dimension, at its intensive equilibrium, which is common to all vertices.

Lemma 1. *On this common realization the descriptor Jacobian is exactly affine, $J(\delta) = J_0 + \sum_i \delta_i J_i$. Each J_i is supported on the states of the devices at bus i and on the two voltage channels of bus i .*

The weights multiply injections only, so the proof is immediate. The spectrum of each vertex equals that of the actual portfolio plus the fillers' -1 eigenvalues. The reduced $A(\delta)$ is not affine, because g_z depends on δ .

B. Factorization

Keep the network variables and eliminate only the device states: $T_S(s) = g_z + g_x(sI - f_x)^{-1}f_z$ evaluated at δ_S . Let E select the $2m$ voltage channels of the candidate buses, let $\Delta Y_i(s) = E_i^\top (T_{\{i\}} - T_0)E_i$ (2×2), $D = \text{blkdiag}(\Delta Y_i)$, and $K = E^\top T_0^{-1}E$. Split $K = K_d + K_o$ into its 2×2 diagonal blocks and the off-diagonal blocks, which carry the network coupling between candidate buses. Set

$$M = DK, \quad Q = (I + DK_d)^{-1}DK_o. \quad (6)$$

Theorem 4 (Network-closure factorization). *Let s be neither a device pole nor a zero of $\det T_0$. Then, for every portfolio S ,*

$$\det P_S(s) = h_S(s) \det T_0(s) \det(I + M_{SS}(s)), \quad (7)$$

$$\det(I + M_{SS}) = \prod_{i \in S} \det(I + M_{ii}) \det(I + Q_{SS}), \quad (8)$$

where P_S is the descriptor pencil and $h_S(s) = \det(sI - f_x)$ is a product of per-device factors. Moreover $\det(I + Q_{SS}) = \sum_{T \subseteq S} \mu_T$ with $\mu_\emptyset = 1$, $\mu_{\{i\}} = 0$, and, for $|T| \geq 2$,

$$\mu_T = \sum_{U: \text{blocks}(U)=T} \det Q_{UU}, \quad (9)$$

the sum of principal minors of Q that touch exactly the blocks of T . μ_T is the Möbius coefficient of $S \mapsto \det(I + Q_{SS})$ and is invariant under block-diagonal changes of port basis.

Proof. The Schur complement of the pencil gives $\det P_S = h_S \det T_S$. By Lemma 1, device blocks couple only through their own bus, so $T_S = T_0 + \sum_{i \in S} E_i \Delta Y_i E_i^\top$ exactly. Sylvester's identity then gives (7). Since $I + DK_d$ is block diagonal, $I + M_{SS} = (I + DK_d)_{SS}(I + Q_{SS})$, which gives (8). The expansion $\det(I + X) = \sum_U \det X_{UU}$ follows from multilinearity; grouping minors by the blocks they touch gives (9). $Q_{ii} = 0$ gives $\mu_{\{i\}} = 0$. Möbius inversion is unique [12]. A change $Q \mapsto S_b^{-1}Q S_b$ with S_b block diagonal leaves every $\det(I + Q_{SS})$, and hence every μ_T , unchanged. $\square$

Corollary 1 (Boundaries are network-closure zeros). *Suppose no member of S alone and no device has an eigenvalue at s^* . Then $s^* \in \sigma(A_\perp(S))$ iff $-1 \in \sigma(M_{SS}(s^*))$, iff $\det(I + Q_{SS}(s^*)) = 0$. If the network coupling is removed ($K_o = 0$), no such s^* exists.*

Thus every non-additive effect of a replacement portfolio is an interaction term μ_T generated by network closure between the replaced buses. The algebraic degree of $S \mapsto \det(I + Q_{SS})$ and κ are different quantities: an affine family can have $\kappa = m$, and a full-degree family can have $\kappa = 1$.

Proposition 4 (Boundary sensitivity). *Let -1 be a simple eigenvalue of $M_{SS}(s^*, \theta^*)$ with right and left eigenvectors q, p . Then the crossing eigenvalue moves as*

$$\frac{\partial s^*}{\partial \theta} = -\frac{p^H \partial_\theta M_{SS} q}{p^H \partial_s M_{SS} q}, \quad (10)$$

and $\nabla_\theta \text{Re } s^*$ is the normal of the boundary.

Proof. Simple-eigenvalue perturbation gives $d\lambda = p^H dM q / p^H q$. The implicit function theorem applied to $\lambda(s, \theta) = -1$ then yields (10). $\square$

C. Symmetry-deflated zero-frequency port

By (5), $\det T_S(0)$ vanishes structurally for every S , so the port test is blind to aperiodic crossings. Let $U = R_x / \|R_x\|$ and $\beta, \beta_2 > 0$. Relocate $f_x^\# = f_x - \beta U U^\top$. The vector $x_0 = w + a R_x$ with $a = \omega_B / \beta - U^\top w / \|R_x\|$ is then a null vector of the relocated reduced matrix. Set $f_x^{\#\#} = f_x^\# - \beta_2 x_0 x_0^\top / \|x_0\|^2$ and $T^{\#\#}(0) = g_z - g_x (f_x^{\#\#})^{-1} f_z$.

Theorem 5 (Zero-frequency port closure). *$\sigma(A^{\#\#}) = \sigma(A_\perp) \cup \{-\beta, -\beta_2\}$ and $\det T^{\#\#}(0) = \det g_z \det A^{\#\#} / \det f_x^{\#\#}$. Along a path on which sign $\det g_z$ and sign $\det f_x^{\#\#}$ are constant, sign $\det T^{\#\#}(0)$ changes iff the number of positive real eigenvalues of $A_\perp$ changes parity.*

Proof. Brauer's rank-one theorem [22] moves one zero to $-\beta$. The Jordan partner becomes the exact null vector x_0 , which a second rank-one update moves to $-\beta_2$. The determinant identity is the Schur formula. Complex pairs contribute $|\lambda|^2 > 0$, so sign $\det A_\perp = (-1)^{\dim A_\perp} (-1)^{\#\{\lambda > 0\}}$. $\square$

The test does not see complex crossings; those are the domain of (7). It also does not see coalescences of two positive real eigenvalues, where nothing crosses $s = 0$.

VI. MONTE CARLO VALIDATION

All pass rules were frozen in a configuration file committed before any draw. One amendment was recorded before the IEEE-39 run: a one-draw smoke test showed that the numerical split of the Jordan pair ($\mathcal{O}(\sqrt{\epsilon\|A\|\omega_B}) \approx 10^{-5}$) had to be excluded from the spectral-identity check. Checks that are logical identities validate the implementation. The informative outputs are the physical checks and the frequencies. Expected-failure controls are declared.

A. Synthetic suites

Each suite draws random systems that satisfy exactly one theorem's hypotheses (Table III).

- **S1** plants the Jordan center in random $A = P \begin{bmatrix} J & X \\ 0 & B \end{bmatrix} P^{-1}$ ($n \in [8, 40]$).
- **S2–S3** use random affine portfolio families.
- **S4** uses Metzler families under random similarities.
- **S5** uses random graphs.
- **S6** uses random block matrices.
- **S7** uses random descriptor systems with a planted center and five classes of spectral events.

S2 quantifies how often the converses of Proposition 2 fail. In 65% of random families some stable target is not safe in every order, and in 4% some stable target cannot be reached by any safe one-at-a-time path. S4 shows that the monotone class is narrow: removing sign structure produces violations in 99.7% of families.

B. IEEE-39 suites

Policies are drawn uniformly from $g = u^2$ ($u \sim U(0, 1)$), $k \in [0.5, 2.3]$, $t \in [0.5, 3]$, $h \in [0, 2]$; portfolios are drawn uniformly from the 16 subsets of the core $\{30, 33, 35, 37\}$.

- **P1** checks Theorem 1 at random equilibria. It also checks the nonlinear symmetries at random off-equilibrium states (10^{-2} perturbation, $\phi \sim U(-\pi, \pi)$).
- **P2** follows random policy segments. Every change of $\mathcal{H}$ is re-bracketed by 40-step bisection, and the crossing eigenvalue is recorded.
- **P3** checks (7) and (9) at random θ and random s .
- **P4** is stratified: 40 draws with $\alpha_{\perp} \leq -0.1$ and 40 with $\alpha_{\perp} \geq 0.1$, each simulated in nonlinear TDS with a 5-MW load pulse.

Fig. 1 summarizes the residuals, the TDS rates, and the crossing frequencies. All preregistered IEEE-39 rules pass. The TDS envelope rate tracks $\alpha_{\perp}$, which confirms that Definition 1 is the stability notion the nonlinear model obeys. The preregistered P2 rule failed as implemented. Its bisection used the three-state label, so it stopped at the edge of the classifier's unresolved band. There the crossing eigenvalue still has $|\operatorname{Re} \lambda| = 4 \times 10^{-6} - 7 \times 10^{-5}$, above the frozen 10^{-6} . A post-hoc re-bisection on the sign of $\alpha_{\perp}$ (same paths, brackets, and threshold) locates all 120 events at $|\operatorname{Re} \lambda| \leq 2 \times 10^{-9}$. Each is a complex pair at 0.41–0.72 Hz with finite $|\lambda|$ and g_z well conditioned along every path, as Theorem 2 requires. We report both results. Across the 2064 fully classified path points, 30 755 targets are stable; 131 of them are not safe in every order, and one has no safe one-at-a-time path.

C. Deviations from the preregistration

Every deviation is listed here and in Table III.

- 1) The P1 spectral-identity rule matched the numerically split Jordan pair. It was amended after a one-draw smoke test and before the run, with the threshold unchanged.
- 2) The P2 rule failed as implemented, because the bisection stopped at the classifier's resolution band. The post-hoc re-bisection is reported beside the failure, not instead of it.
- 3) In S7, three draws of the “coalescence” class contained a genuine zero crossing, because the generator allowed $a - d < 0$. The port flipped in exactly those draws. 183 of 1000 S7 draws were left undecided by the frozen device-flip rule, because synthetic device blocks are not block diagonal; in the Kundur holdout no bracket was undecided.
- 4) Two P1 draws needed a redrawn off-equilibrium state because the first one left the algebraic chart.

In the second-order audit, the first run used a first-order-hold coefficient that was $1/\Delta t$ too large. The error was found before any interpretation and corrected; only the corrected run is reported.

VII. IEEE 39-BUS RESULTS

A. Benchmark

The IEEE 39-bus system [23] is taken from the ANDES dynamic case [24], [25], with ten two-axis SGs; bus 39 represents the interconnection and is never replaced. The documented IEEEEX1 exciter is harmonized into a first-order AVR (gain of the regulator, time constant of the exciter), a declared model change. The core candidates are SGs 30, 33, 35, 37. Replacing all four displaces 2096.6 MW of active dispatch on 4270.7 MVA of converter rating. Every result in this section is conditional on this model.

B. Policy dependence of the witness structure

Three policy planes were mapped: F7A ($g \times k$), F7B ($g \times t$), and F7C ($g \times h$). All 345 229 points were re-audited in transverse coordinates on the direct path. No resolved label changed relative to the frozen maps; 1.5% of points remain within the classifier's numerical resolution. The planes carry 30, 36, and 16 distinct hypergraphs, with up to 15 hypergraphs behind a single value of κ (Fig. 3).

At the reference point P4 ($g = 0.03625$, $k = 1.425$, $t = 1.5$, $h = 1$), $\mathcal{H} = \{\{30, 33, 35, 37\}\}$ and $\alpha_{\perp} = +0.127 \text{ s}^{-1}$: every proper subset is stable and the four-unit portfolio is not. At full voltage-support gain (P_{∞} : $g = 1$, $k = 0.5$), $\mathcal{H} = \emptyset$. On the pure-policy line $t = 0.852$ the minimal witness contracts $4 \rightarrow 3 \rightarrow 2$ and re-expands $3 \rightarrow 4 \rightarrow \emptyset$ as g increases.

The failing mode is the 0.64–0.71 Hz inter-area family, not the amount of dispatch removed (Fig. 2). Among the 512-portfolio census, consider the 25 stable four-unit portfolios whose displaced active dispatch matches that of the 10 unstable ones. In all 25 the family stays damped ($\max \operatorname{Re} \leq -0.114 \text{ s}^{-1}$, $\zeta \geq 3.9\%$). It is unstable in 8 of the 10 failing portfolios; the other two fail through a different mode. Total

TABLE III
PREREGISTERED MONTE CARLO VALIDATION OF EVERY STATEMENT

Statement	Suite	Draws	N	Measured	Frozen rule / result
Thm 1 (linear)	S1 synthetic	1000	1000	exact RHP count 1000/1000; (5) $\leq 5.5 \times 10^{-9}$; basis 1.5×10^{-16}	$\leq 10^{-8}$: pass
control: magnitude cutoff	S1	1000	1000	wrong in 235/235 positive planted modes, 0/765 otherwise	expected failure: confirmed
Props 1,2	S2 synthetic	2000	30881 targets	0 violations; A\B in 86 families, B\C in 1276, non-hereditary 1296	0 violations: pass
Thm 2	S3 synthetic	300	1044 crossings	every $\mathcal{H}$ change bracketed (698 real, 346 complex)	0 exceptions: pass
Thm 3	S4 synthetic	1000	80 000 edges	0 non-monotone edges, A=B=C	pass
control: signs removed	S4	1000	80 000	37 166 non-monotone edges in 997/1000 families	expected failure: confirmed
Prop 3	S5 synthetic	500	exhaustive	0 mismatches with k -cliques (310 graphs contain one)	pass
Thm 4 (algebra)	S6 synthetic	500	500	minors 3.4×10^{-15} , Möbius 1.3×10^{-15} , invariance 8.2×10^{-12}	$\leq 10^{-10}$: pass
Thm 5	S7 synthetic	1000	817 decided	flip $\Leftrightarrow$ parity change in 817/817; 183 undecided by the device-flip rule	pass ^a
Thm 1 (physical)	P1 IEEE-39	300	300	dim $C = 2$ in 300/300; spectrum 4.5×10^{-9} ; Jordan entry 3.9×10^{-12}	pass
Thm 1 (nonlinear)	P1 IEEE-39	300	300	$r(x + \phi R_x)$: 2.4×10^{-14} ; $r(x + cw)$: 9.4×10^{-15}	$\leq 10^{-8}$: pass
Thm 2, Prop 2	P2 IEEE-39	120 paths	120 events	101 changes of $\mathcal{H}$ on 28 paths; all crossings complex, 0.41–0.72 Hz; cond(g_z) $\leq 2.8 \times 10^3$	fail as implemented^c ; post hoc: 120/120
Thm 4 (physical)	P3 IEEE-39	40	120	ratio 3.2×10^{-13} ; Möbius 8.1×10^{-13}	$\leq 10^{-8}$: pass
Def. 1 in TDS	P4 IEEE-39	80	80	TDS verdict = sign of $\alpha_\perp$ in 80/80; rate vs. $\alpha_\perp$: $r = 0.969$	80/80: pass
Thm 5 (physical)	Kundur holdout	474	29+445	28/29 detected; 0/445 false positives	miss = coalescence ^b

^aThree draws of the “coalescence” class contained a genuine zero crossing through a generator defect; the port flipped in exactly those. ^bTwo positive real eigenvalues merge into a complex pair; no eigenvalue passes through $s = 0$ (Theorem 5). ^cThe bisection used the three-state label and stopped at the classifier’s resolution band ($|\text{Re } \lambda| \leq 7 \times 10^{-5} > 10^{-6}$); re-bisection on the sign of $\alpha_\perp$ (post hoc, threshold unchanged) gives $\leq 2 \times 10^{-9}$.

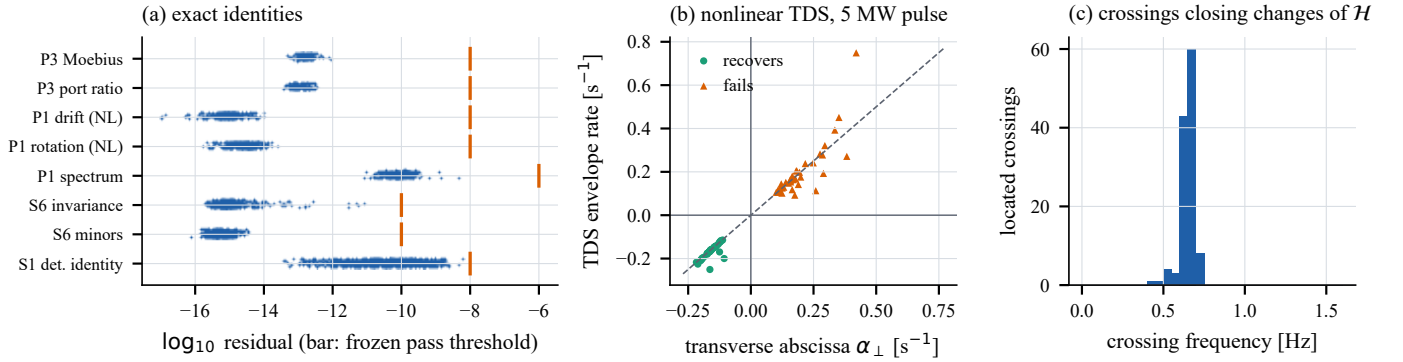


Fig. 1. Monte Carlo validation. (a) Residuals of the exact identities (S1, S6 synthetic; P1, P3 IEEE-39) against their frozen thresholds (bars). (b) Nonlinear TDS envelope growth rate versus the transverse abscissa for 80 random IEEE-39 policies and portfolios (dashed: identity). (c) Frequencies of the transverse eigenvalues that close the 120 changes of $\mathcal{H}$ on 120 random policy paths (post-hoc re-bisection; see Table III, note c).

displaced dispatch predicts the minimal coalition poorly (AUC 0.57).

C. Network-closure anatomy of the boundaries

Ten frozen boundaries were re-located and analyzed with Theorem 4: the nine witness changes of the F7B line $t = 0.852$ and the flagship boundary on the P4 line ($g^* = 0.2077$, 0.706 Hz).

- The descriptor blocks f_x, f_z, g_x, g_z are affine in δ to 4.5×10^{-16} (order- ≥ 2 Möbius coefficients). Each replacement touches only its own devices and bus (a 2×2 g_z block).

- (7) holds to 1.4×10^{-13} for all 16 subsets at all ten points, and the critical eigenvalue of M_{SS} equals -1 to 8.5×10^{-14} .
- The local factor $\prod_i \det(I + M_{ii})$ stays between 0.08 and 0.31. The zero is carried entirely by $\det(I + Q_{SS})$ (Fig. 5).
- No truncation of (9) below order $|S|$ vanishes: the residual is 0.29–0.40 for triples at order 2, and 0.59–0.60 (order 2) and 0.13–0.16 (order 3) for the four-unit portfolio.

Every observed boundary is therefore closed only by the full-order network interaction of the coalition whose label changes, which is the mechanism behind κ . Table V lists the ten events and Fig. 4 places them on the F7B line.

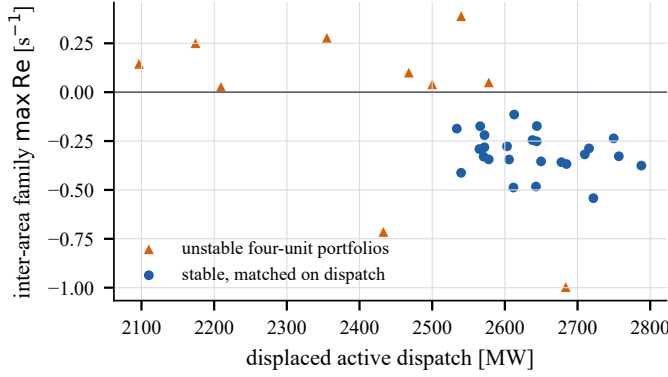


Fig. 2. Inter-area family abscissa versus displaced active dispatch for the unstable four-unit census portfolios and 25 stable portfolios matched on dispatch. Instability is not a matter of how much dispatch is displaced.

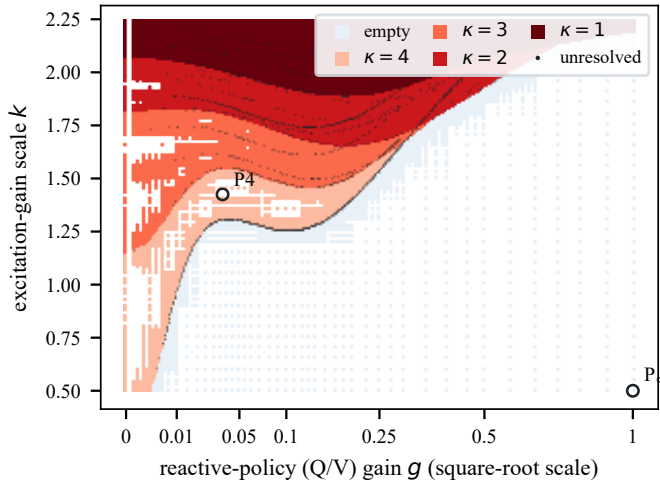


Fig. 3. Transverse minimum incompatible order κ over the F7A policy plane of IEEE-39 ($t = 1.5$, $h = 1$, core 30/33/35/37, matched dispatch, no governor). Black dots: points within classifier resolution.

Table IV gives the build-up at the flagship boundary, order by order. The interaction coefficients of orders 2, 3, and 4 have comparable magnitude (0.51, 0.45, 0.16). The fourth-order term, a sum of principal minors of Q touching all four replaced buses, is exactly what closes the determinant. On 120 random (θ, s) draws (suite P3), the full-order coefficient of $\det(I + M_{SS})$ ranges from 7.8×10^{-4} to 469 and never vanishes. No lower-degree representation of the portfolio family is exact on this benchmark.

Two negative checks bound this statement. (i) The effective rank of Q (threshold 10^{-2}) stays at 7–8 of 8 throughout $\kappa = 4 \rightarrow 3 \rightarrow 2 \rightarrow 3 \rightarrow 4$, so interaction dimension does not explain the contraction. What varies is the magnitude: $\sigma_1(Q)$ is 6.1–8.9 at the contracting events and 1.7–1.8 at the four-unit boundaries (descriptive only). (ii) A path (walk) expansion of T_0^{-1} with bus-diagonal splitting diverges: $\rho(D_g^{-1}N) = 1.003$ – 1.556 over 0.3–1.5 Hz, both at the flagship boundary and for a matched stable control. No path interpretation is offered.

TABLE IV
ORDER-BY-ORDER EXPANSION AT THE FLAGSHIP BOUNDARY (P4 LINE, 0.706 Hz)

Order k	$\det(I + M_{SS})$		$\det(I + Q_{SS})$	
	$ \sum_{ T =k} \mu_T $	$ \text{trunc.} \leq k $	$ \sum_{ T =k} \tilde{\mu}_T $	$ \text{trunc.} \leq k $
0	1	1	1	1
1	2.509	1.712	0	1
2	2.393	0.889	0.509	0.605
3	0.989	0.146	0.445	0.161
4	0.146	7×10^{-15}	0.161	8×10^{-14}

The left pair includes the local terms; in the right pair they are factored out ($\tilde{\mu}_{\{i\}} = 0$, Theorem 4).

TABLE V
NETWORK-CLOSURE ANATOMY OF TEN FROZEN IEEE-39 BOUNDARIES

Ev.	S changing	g^*	Hz	local	trunc. ^a	$\sigma_1(Q)$	rank ^b	angle ^c
E1	30+33+35	0.0102	0.628	0.315	0.336	6.13	7	2.3e-4
E2	30+33+37	0.0188	0.633	0.198	0.396	6.88	7	7.1e-5
E3	33+35+37	0.0609	0.671	0.080	0.385	4.00	8	2.7e-4
E4	30+33	0.1034	0.656	0.081	1	8.93	7	1.8e-4
E5	33+35+37	0.1870	0.695	0.135	0.296	2.37	8	3.3e-4
E6	30+33	0.2491	0.672	0.104	1	6.06	7	8.4e-5
E7	30+33+35	0.2778	0.706	0.130	0.372	2.01	8	3.5e-4
E8	30+33+37	0.2931	0.685	0.132	0.293	3.46	8	2.0e-4
E9	30+33+35+37	0.2973	0.718	0.134	0.132	1.67	8	2.9e-4
F	30+33+35+37	0.2077	0.706	0.089	0.161	1.84	8	4.3e-4

E1–E9: F7B line $t = 0.852$ ($k = h = 1$); F: P4 line ($k = 1.425$, $t = 1.5$).

^a $|\det(I + Q_{SS})|$ truncated at order $|S| - 1$ (the full order gives $\leq 4 \times 10^{-13}$). ^bEffective rank of Q at 10^{-2} . ^cBoundary-normal error of (10) versus direct finite differences [deg].

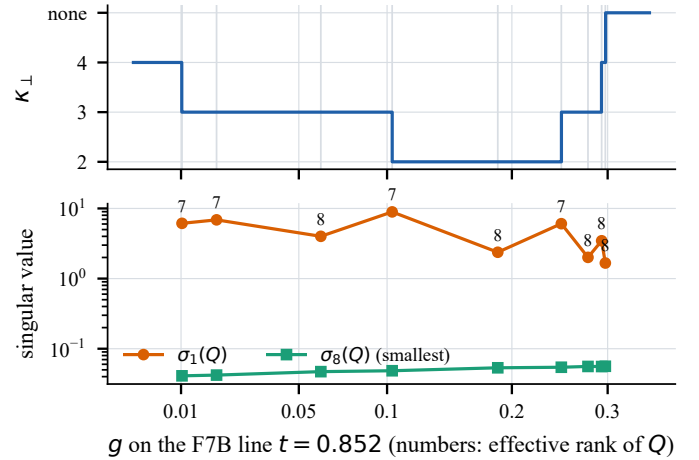


Fig. 4. F7B policy line $t = 0.852$. Top: the transverse minimum incompatible order contracts $4 \rightarrow 3 \rightarrow 2$ and re-expands. Bottom: largest and smallest singular values of the interaction operator Q at the nine boundary events (labels: effective rank). The dimension stays full while κ changes.

D. Boundary motion and retuning

Formula (10) was compared with direct finite differences of the tracked transverse eigenvalue in all four policy coordinates. At all ten boundaries, the boundary normal agrees to 4.3×10^{-4} degrees, its magnitude to 4.3×10^{-6} (relative), and the frequency drift to 9×10^{-6} .

Consider making the P4 portfolio stable by a pure Q/V-gain change.

- A single linear step predicts $\Delta g = 0.052$, whereas

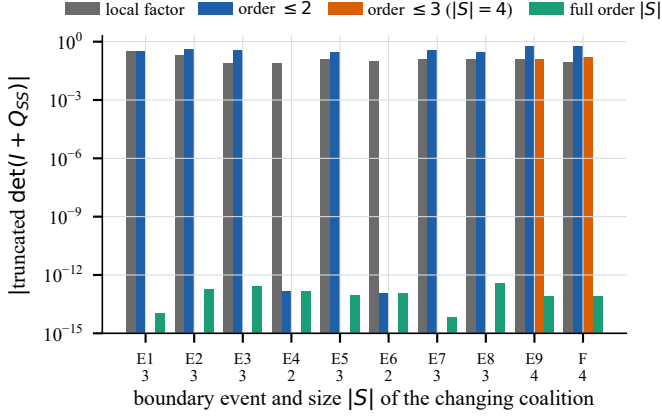


Fig. 5. Anatomy of ten frozen IEEE-39 boundaries (E1–E9: F7B line $t = 0.852$; F: flagship on the P4 line). Local factor and truncations of the interaction expansion (9). Only the full-order term (order $|S|$) reaches zero. For pairs, order ≤ 2 is the full order.

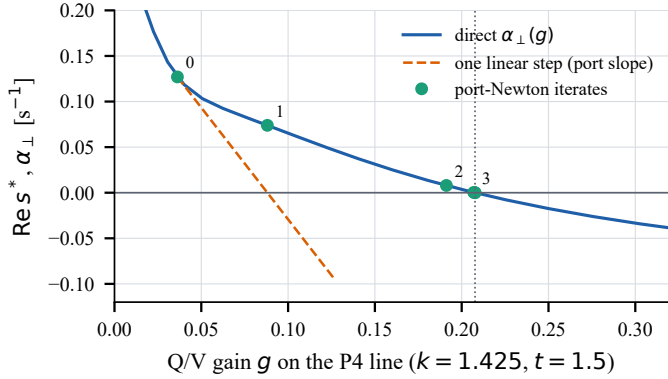


Fig. 6. Q/V retuning of the P4 four-unit portfolio. The single linear step built from the port derivative (10) undershoots, because the slope changes by a factor of five along the path. Newton iteration with the same derivative reaches the continuation boundary (dotted) to 2×10^{-10} .

continuation gives 0.171: $\partial \text{Re } s / \partial g$ drifts from -2.45 at P4 to -0.46 at the boundary.

- Newton iteration with (10) reaches the continuation boundary $g^* = 0.20768$ to 2×10^{-10} in five steps.

The port derivative is thus an exact local tool, and an iterative one for retuning (Fig. 6).

E. Robustness to documented primary frequency control

The same source workbook documents TGOV1N turbine governors ($R = 0.05$, $T_1 = 0.05$, $T_2 = 1$, $T_3 = 2.1$ s), which were added untuned as a separately versioned model. With governors, $C = \text{span}\{R_x\}$ (Theorem 1). Results (Fig. 7):

- The P4 four-unit portfolio becomes stable ($\alpha_{\perp} = -0.0745$), although it remains the least-damped portfolio of the lattice.
- On a 399-point plane the number of distinct hypergraphs falls from 26 to 10. Non-empty structure remains at 74 points ($\kappa = 2, 3, 4$ at 18, 32, and 24 points); no singleton ever fails, and the base is always stable.

TABLE VI
FROZEN VERSUS GOVERNED IEEE-39 (DOCUMENTED TGOV1N)

Quantity	Frozen	Governed
P4 four-unit $\alpha_{\perp}$	+0.127	−0.0745
P4 line, $\kappa = 4$ window	$g \in [0.016, 0.20]$	$g \in [0.016, 0.023]$
tongue line $k = 1.30$	U–S–U–S (re-entry)	no re-entry
plane: distinct $\mathcal{H}$	26	10
plane: non-empty points	148 (+21 base unstable)	74 (base always stable)
plane: $\kappa = 1$ points	48	0
reference points kept	5 of 6 non-empty	P3 only
damped condenser, P4	threshold 2.48–3%	not needed

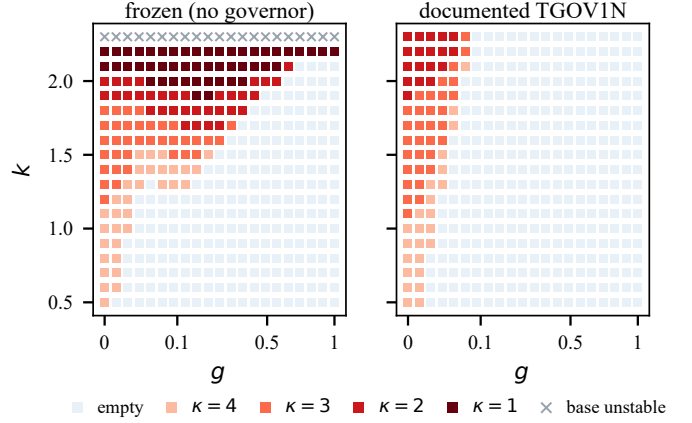


Fig. 7. Minimum incompatible order on a coarse IEEE-39 policy plane ($t = 1.5$, $h = 1$), frozen model versus the model with the source’s documented TGOV1N governors.

- Of the six reference points, one (P3) keeps its structure ($\{30, 33, 35\}$, $\{30, 33, 37\}$); P1, P2, P4, and P_{fold} become empty.
- Independently, in ANDES the governors move the base inter-area pair from -0.222 to -0.770 s^{-1} , consistent with the damping direction.

Policy-dependent incompatibility therefore survives governors in a smaller region, while the specific four-unit coalition at P4 is conditional on the absence of primary frequency control. Table VI compares the two models on the P4 line and on the tongue line $k = 1.30$. With governors the tongue disappears, and the four-unit window on the P4 line shrinks from $g \in [0.016, 0.20]$ to $g \in [0.016, 0.023]$.

F. Nonlinear confirmation and model scope

In earlier preregistered runs, all 32 declared small-disturbance TDS verdicts agree with the eigenvalues (frequency within 2.5×10^{-4} Hz, damping within 6×10^{-4} s^{-1}), and P4 extends this to 80 random cases.

For finite disturbances we used three frozen families: a bus-16 load pulse (D1), an SG36 mechanical-power dip (D2), and a GFL output dip (D3). We also declared model-scope guards: GFL terminal voltage in $[0.9, 1.1]$ pu, GFL current ≤ 1.2 pu, PLL and COI frequency bands, and the documented exciter and stabilizer limits. Findings:

- Inside these guards, the finite-disturbance hypergraph equals $\mathcal{H}(\theta)$ for all six (point, family) combinations.

TABLE VII
FIRST LYAPUNOV COEFFICIENT AT THREE IEEE-39 BOUNDARIES

Boundary	p^*	Hz	$\partial\alpha_{\perp}/\partial p$	ℓ_1	type
four-unit, P4 line ($p = g$)	0.2077	0.706	-0.461	+0.0095	subcrit.
tongue, $k = 1.30$ ($p = g$)	0.0509	0.639	+1.102	+0.0250	subcrit.
damped condenser (p : rating)	2.48%	0.621	-5.570	+0.026	subcrit.

ℓ_1 from polarized central differences of the exact transverse field at two step sizes (spread $\leq 0.2\%$); transversality holds at all three points.

- No transversely stable portfolio fails inside the guards: 84 of 96 thresholds are guard activations (GFL voltage band, stabilizer limit).
- Normal-form analysis on the exact transverse system shows that the three examined oscillatory boundaries are subcritical Hopf points [26] (Table VII).

Nonlinear composability is therefore not stricter than transverse spectral composability within the model's validity. Beyond the guards the model does not decide.

Fig. 8 shows the resilience radii $R_k = \min_{|S| \leq k} r_S$, where r_S is the largest tolerated severity. At P4 the model-scope radius is $\rho_{\text{scope}} = 130.5$ MW (D1), 97.7 MW (D2), and 0.091 (D3). At P_{∞} it is 88.3 MW, 360.8 MW, and 0.308. The reactive policy that repairs small-signal composability therefore widens two envelopes and narrows one.

The order in which portfolios first reach the envelope is itself structured. At P4, D1, the four-unit portfolio is first, then 30+33+35, then pairs, then singletons, and finally the unreplaced base at 681 MW. This first-event structure records where omitted limiters would act; it says nothing about stability beyond that point.

A second-order audit on the same DAE (Fig. 9) confirms that the algebraic curvature $f_z D^2 \psi$ must be retained [27]. At 2 MW, the full second-order response is 170–450 times more accurate than the linear one in frequency and 750–3400 times in bus voltage (cubic remainder, slope 3.0–3.2). Dropping $f_z D^2 \psi$ breaks the rotation/drift symmetry at second order and makes the frequency prediction 570–1700 times worse than linear. With the center-subspace component removed, the device curvature alone is still no better than linear.

G. Planning consequences

Proposition 2 turns $\mathcal{H}$ into constraints. On the 512 portfolios of nine candidates (30–38) at one reference policy, 327 are transversely stable, 324 are safe in every order, and three are final-stable but order-dependent. In each of the three, the only unstable subset is the four-set without unit 34, so exactly the orders that convert 34 last are unsafe. These three pairs also exclude the monotone class (Theorem 3).

The MW-optimal plan (31+32+34+35+37+38, 3652.3 MW / 6499.9 MVA) is safe in any order. At P4 the optimal any-order-safe core plan is 30+33+35 (1775.1 MW); at P_{∞} it is the full core (2096.6 MW).

A condenser with a damped swing mode restores P4 above 2.48% of the retired rating (106.0 MVA; TDS threshold 2.47%), while an undamped low-inertia condenser is itself unstable. Keeping a 200-MW disturbance inside the declared

TABLE VIII
PLANNING RESULTS ON IEEE-39 (MW AND MVA ON SEPARATE AXES)

Problem	Setting	Result
P1 any-order safe	core, P4	30+33+35: 1775.1 MW / 3300.5 MVA
P1 any-order safe	core, P_{∞}	30+33+35+37: 2096.6 MW / 4270.7 MVA
P1 any-order safe	9-unit census	31+32+34+35+37+38: 3652.3 MW / 6499.9 MVA
Order dependence	9-unit census	3 of 327 stable targets (34 must not be last)
P2 nonlinear safe	core, P4	out of model scope ($\rho^* \geq \rho_{\text{scope}}$, 3 families)
P2 nonlinear safe	core, P_{∞} , D2	identical to P1
P3 support	P4, small signal	2.48% = 106.0 MVA (damped condenser)
P3 support	P4, envelope	792.4 / 684.0 MVA; D3 not reached at 1067.7 MVA
Governors	D1/D2	stable without support
	P4	($\alpha_{\perp} = -0.0745$)

converter voltage band requires 684–792 MVA. That is an envelope requirement of this model, not a stability requirement. Table VIII summarizes the three planning problems, preregistered before any threshold was computed:

- P1 maximizes displaced active dispatch subject to containing no hyperedge.
- P2 adds the nonlinear hyperedges at preregistered severities (200 MW for D1 and D2, 0.35 for D3).
- P3 minimizes the damped-condenser rating.

VIII. DISCUSSION

A. Relation to existing methods

Table IX places the object among established method families. The hypergraph does not replace any of them. Its per-portfolio labels *are* generalized Nyquist tests. What it adds is the organization of those labels over binary replacement decisions and their dependence on the converter policy, which none of the configuration-level criteria resolves. Flagship-only Nyquist, single-port impedance, and short-circuit-ratio screens miss the coalition changes on IEEE-39. Modal-sensitivity, additive, and pairwise predictors of the four-unit instability were correct in 22, 30, and 38 of 52 cases, respectively.

B. Use in retirement planning

Three uses follow directly.

- 1) The any-order-safe portfolios are exactly those avoiding every hyperedge (Proposition 2). This is a set-packing constraint that a unit-commitment or capacity-expansion model can carry.
- 2) Order dependence is real but rare on IEEE-39 (three of 327 stable targets at one policy, 131 of 30755 on random paths). Where it occurs, the unsafe orders are identified exactly.
- 3) The policy is a planning variable. At fixed network and dispatch, it moves κ between 1 and ∞ , and the boundary normals given by (10) are exact local directions for retuning.

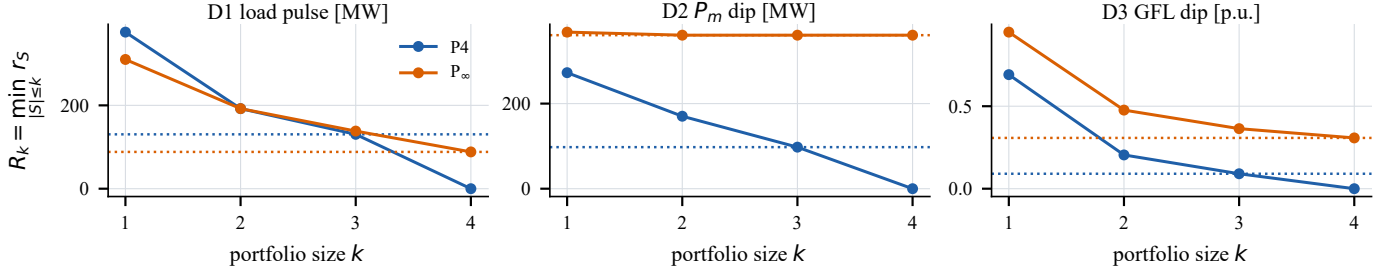


Fig. 8. Finite-disturbance resilience radii R_k of the IEEE-39 core at P4 and P_∞ for three frozen disturbance families (TDS, phasor DAE). Dotted: model-scope radius ρ_{scope} . Below it, the nonlinear hypergraph equals $\mathcal{H}(\theta)$; above it, every threshold is a guard activation.

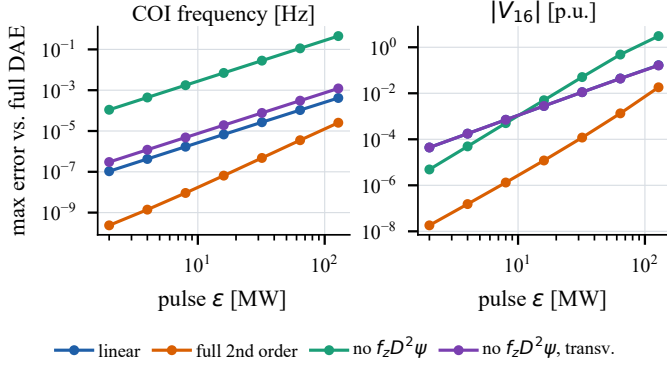


Fig. 9. Second-order reduced response of the critical triple 30+33+35 at P4 against the full DAE (bus-16 load pulse). The complete second-order model requires the algebraic curvature $f_z D^2 \psi$.

TABLE IX
RELATION TO ESTABLISHED METHOD FAMILIES

Family	Relation to this work
Generalized Nyquist [10]	per-portfolio test; our labels are GN tests
Impedance / SCR [4], [7]	configuration screens; miss coalition changes
Small gain, passivity	our sampled small-gain screen fails ($\rho = 2.9\text{--}12.3$)
μ / robust stability [19]	NP-hard in general; Prop. 3 adds no priority
Sparse stability radius [21]	continuous structured perturbations, not binary replacement
$N-k$ minimal sets [14]	static analogue; ours is dynamic, transverse, policy dependent
Selective modal analysis [28]	explains modes; does not enumerate coalitions
Continuation / Hopf [26]	boundaries are subcritical Hopf points
Energy / Lyapunov	local bound $\leq 10^{-11}$ of the measured threshold
Higher-order modal [27]	requires $f_z D^2 \psi$ (Fig. 9)

Evaluating 2^m labels is exponential in general (Proposition 3); with $m = 4$ it takes 16 evaluations. The factorization in (7) evaluates all portfolios from one base network and m local increments at a common operating point.

IX. CORROBORATION AND LIMITATIONS

Kundur two-area system. Policy dependence recurs: 61/61 and 36/36 policy lines change $\mathcal{H}$, which is never empty. All

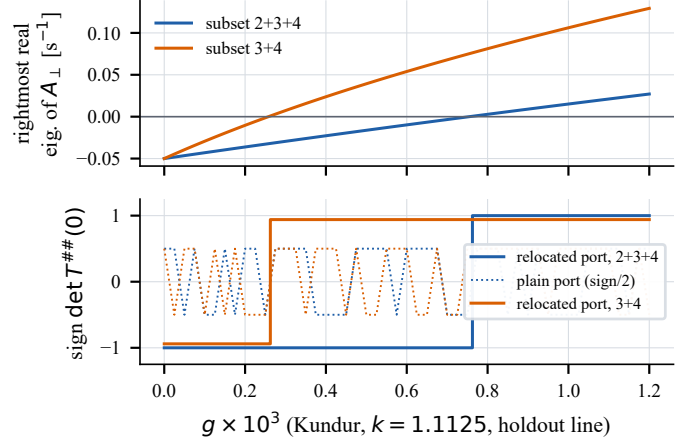


Fig. 10. Symmetry-deflated zero-frequency port on a Kundur holdout line. Top: the physical real transverse eigenvalue crosses zero. Bottom: the relocated port changes sign exactly at the crossings, whereas the plain port (dotted) is structurally singular and its sign is numerical noise.

194 aperiodic (Q/V-integrator) crossings are real transverse crossings. They motivated Theorem 5; its preregistered holdout (Fig. 10) is the physical validation, whereas the 194 design crossings are not counted.

IEEE 68-bus system [29]. The documented model is reproduced (15/15 modes to 5×10^{-4} Hz), but the preregistered four-candidate policy map is empty. Policy dependence appears only at ≥ 6 of 12 replaced plants ($\kappa = 6 \rightarrow 9 \rightarrow 11$). The structure is therefore system dependent, not universal.

Independent implementation. ANDES reproduces the power flow, the base inter-area mode (within 4.1%), and the governor damping direction. ANDES does not implement the converter model used here, so no portfolio result is claimed as independently reproduced.

Model scope. The model is a phasor DAE, not an electromagnetic transient model. It has no converter current limits, ride-through logic, or DC link. The frozen models have no primary frequency restoration, so all stability statements are transverse. No certificate is claimed: sampled small-gain and principal-minor screens give $\rho = 2.9\text{--}12.3 > 1$, and a local Lyapunov bound reaches at most 10^{-11} of the measured threshold.

X. CONCLUSION

The minimal incompatibility hypergraph is a planning object for SG retirement: it is exactly defined in transverse coordinates, it changes only at imaginary-axis crossings, and it characterizes the portfolios that are safe in every implementation order. Its non-additivity is a principal-minor expansion of the network closure between replaced buses. On IEEE-39 every observed boundary is closed only by the full-order interaction of the changing coalition, and the converter reactive policy changes which coalitions fail. Every statement was tested by preregistered Monte Carlo suites, synthetic and physical. All passed except one IEEE-39 localization rule, which failed as implemented; that failure is reported together with its post-hoc correction. The four-unit coalition is conditional on the absence of primary frequency control. Nonlinear composability adds no stricter structure within model scope, and the structure is system dependent. Extending the analysis to limiter-equipped, governed, and mixed grid-forming benchmarks requires new, separately versioned models.

APPENDIX A PROOF DETAILS

Nonlinear drift in Theorem 1. Fix x in the chart domain and let $z = \psi(x)$. Consider $x_c = x + cw$. Speeds enter the reduced equations only through $\dot{\delta} = \omega_B(\omega - 1)$ and, for a GFL, through $\dot{\vartheta} = k_{p,\text{pll}}v_q + x_{\text{pll}}$. In both cases the shift adds the constant $c\omega_B$ to the angle derivatives, and nothing else. The power-form swing equation with $D = 0$ does not contain ω on its right-hand side, the stabilizer uses electrical power, and the network is frequency independent. Hence $\psi(x_c) = \psi(x)$ and $r(x_c) = r(x) + c\omega_B R_x$. The rotation identity follows from $\psi(x + \phi R_x) = e^{j\phi}\psi(x)$ and the invariance of f under a common rotation of angles and voltages. Both identities hold for every x , so they are properties of the vector field, not only of the linearization.

Locality in Theorem 4. Order the device states by bus. On the common realization, f_x is block diagonal by device; f_z and g_x couple a device only to the two channels of its own bus; and the device contribution to g_z is a 2×2 block on that bus. Therefore $g_x(sI - f_x)^{-1}f_z = \sum_{\text{devices}} E_b G_b(s) E_b^\top$, where G_b is the device's 2×2 admittance. With the network and load terms common to all vertices, $T_S - T_0 = \sum_{i \in S} E_i \Delta Y_i E_i^\top$. The increment ΔY_i does not depend on S because every device stays at its common intensive equilibrium (matched dispatch). The residual of this locality on the direct (non-common) path is bounded by the power-flow tolerance and was measured below 8×10^{-8} .

Corollary 1. If member i alone has no eigenvalue at s^* , then $\det(I + M_{ii}) = \det T_{\{i\}}(s^*) / \det T_0(s^*) \neq 0$. The zero of (8) must then be a zero of $\det(I + Q_{SS})$. With $K_o = 0$, $Q = 0$ and $\det(I + Q_{SS}) = 1$.

APPENDIX B MONTE CARLO CONSTRUCTIONS

S1. $n \sim U\{8, \dots, 40\}$. The transverse spectrum mixes real eigenvalues and complex pairs with real parts in $[-2, 1]$ ($|\text{Re}| \geq 0.01$) and imaginary parts in $[0.5, 10]$. In odd draws

one real eigenvalue is planted at $\pm u$, with u log-uniform in $[10^{-5}, 5 \times 10^{-4}]$. $P = Q(I + 0.3G)$ with Q Haar orthogonal. The identity (5) is evaluated at three random s with $|s| \in [0.1, 10]$.

S2. $A(\delta) = A_0 + \sum \delta_i A_i$ with $m \sim U\{4, 5, 6\}$, $n = 10$, $\alpha(A_0) \in [-1, -0.2]$, and rank-2 actions scaled by $U(0.3, 1.5)$. Classes A, B, and C are computed by exhaustive lattice search.

S3. The same families with actions $A_i + pB_i$, $p \in [0, 1]$ on a 401-point grid; each change of $\mathcal{H}$ is bisected in 50 steps.

S4. $B(S) = B_0 + \sum_{i \in S} N_i$ with B_0 Metzler, $\alpha(B_0) \in [-2, -0.5]$, and $N_i \geq 0$ at density 0.3, under random similarity ($\text{cond} \leq 10^3$). The control uses random signs.

S5. $G(n, p)$ graphs with $n \in [7, 10]$, $p \in [0.3, 0.8]$, $k \in \{3, 4, 5\}$; all $|S| \leq k$ are enumerated.

S6. Complex Gaussian M with $m \in \{3, 4, 5\}$ blocks of size 2 and random block basis changes.

S7. Random descriptors with $n_x \in [10, 30]$, $n_z \in [6, 20]$, and $f_x = A + f_z g_z^{-1} g_x$, where A is planted as in S1 at both ends of a bracket. Five event classes: single real crossing, double real crossing, real coalescence, complex crossing, and none. Relocation uses $\beta = \beta_2 = 1$ with the consistency pair (2, 0.5). The frozen rule leaves a bracket undecided when $\text{sign det } f_x^{\#\#}$ flips.

P1–P4. Policies are sampled as in Section VI. Each case is solved on the direct path (full AC equilibrium and central-difference Jacobians); P3 uses the common realization of Lemma 1. The random seed and every pass rule are fixed in the preregistration file.

APPENDIX C NONLINEAR SIMULATION DEFINITIONS

The TDS integrates the full DAE (BDF, relative tolerance 10^{-7} , maximum step 0.05 s). The network equations are solved by Newton at every step.

- *Pulse.* Each disturbance is applied for 0.5 s and followed by 30 s of free response.
- *Deviation measure.* $D(t)$ is the larger of the transverse state and voltage deviations, normalized by 10^{-3} and 10^{-2} .
- *Recovers.* D decays below 10% of its early peak with a negative fitted envelope rate.
- *Fails.* D exceeds twice the early peak with positive growth over the last 10 s, or the relative rotor angle exceeds π .
- *Outside model scope.* A guard trips: bus voltage outside $[0.8, 1.2]$, GFL terminal voltage outside $[0.9, 1.1]$, GFL current or apparent power above 1.2, reactive power above 1.0, PLL or speed frequency outside $[-3, 1.8]$ Hz, COI frequency beyond ± 0.5 Hz, field voltage outside the documented exciter limits, or stabilizer output beyond 0.1 (all per unit on the device base).

Thresholds r_S are obtained by continuation on a frozen amplitude grid followed by bisection. A threshold whose first event is a guard is scope-censored, and the minimum of the censored thresholds defines ρ_{scope} .

REPRODUCIBILITY

Code, frozen configurations, the preregistration and its amendment, the Monte Carlo draws, and the source data of every figure are available from the author and in the project repository.

REFERENCES

- [1] N. Hatziaargyriou, J. Milanović, C. Rahmann, V. Ajjarapu, C. Cañizares, I. Erlich, D. Hill, I. Hiskens, I. Kamwa, B. Pal, P. Pourbeik, J. Sanchez-Gasca, A. Stanković, T. Van Cutsem, V. Vittal, and C. Vournas, "Definition and classification of power system stability – revisited & extended," *IEEE Transactions on Power Systems*, vol. 36, no. 4, pp. 3271–3281, 2021.
- [2] F. Milano, F. Dörfler, G. Hug, D. J. Hill, and G. Verbič, "Foundations and challenges of low-inertia systems," in *Power Systems Computation Conference (PSCC)*, Dublin, Ireland, 2018.
- [3] U. Markovic, O. Stanojev, P. Aristidou, E. Vrettos, D. Callaway, and G. Hug, "Understanding small-signal stability of low-inertia systems," *IEEE Transactions on Power Systems*, vol. 36, no. 5, pp. 3997–4017, 2021.
- [4] J. Sun, "Impedance-based stability criterion for grid-connected inverters," *IEEE Transactions on Power Electronics*, vol. 26, no. 11, pp. 3075–3078, 2011.
- [5] L. Harnefors, M. Bongiorno, and S. Lundberg, "Input-admittance calculation and shaping for controlled voltage-source converters," *IEEE Transactions on Industrial Electronics*, vol. 54, no. 6, pp. 3323–3334, 2007.
- [6] X. Wang and F. Blaabjerg, "Harmonic stability in power electronic-based power systems: Concept, modeling, and analysis," *IEEE Transactions on Smart Grid*, vol. 10, no. 3, pp. 2858–2870, 2019.
- [7] H. Xin, W. Dong, X. Yuan, D. Gan, K. Wang, and H. Xie, "Generalized short circuit ratio for multi power electronic based devices infeed to power systems," *Proceedings of the CSEE*, vol. 36, no. 22, pp. 6013–6027, 2016.
- [8] North American Electric Reliability Corporation, "Integrating inverter-based resources into low short circuit strength systems," NERC, Reliability Guideline, Dec. 2017.
- [9] A. G. J. MacFarlane, "Return-difference and return-ratio matrices and their use in analysis and design of multivariable feedback control systems," *Proceedings of the Institution of Electrical Engineers*, vol. 117, no. 10, pp. 2037–2049, 1970.
- [10] A. G. J. MacFarlane and I. Postlethwaite, "The generalized Nyquist stability criterion and multivariable root loci," *International Journal of Control*, vol. 25, no. 1, pp. 81–127, 1977.
- [11] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed. Cambridge University Press, 2012.
- [12] G.-C. Rota, "On the foundations of combinatorial theory I. Theory of Möbius functions," *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete*, vol. 2, no. 4, pp. 340–368, 1964.
- [13] R. P. Stanley, *Combinatorics and Commutative Algebra*, 2nd ed. Boston: Birkhäuser, 1996.
- [14] D. Bienstock and A. Verma, "The $N - k$ problem in power grids: New models, formulations, and numerical experiments," *SIAM Journal on Optimization*, vol. 20, no. 5, pp. 2352–2380, 2010.
- [15] L. Farina and S. Rinaldi, *Positive Linear Systems: Theory and Applications*. New York: Wiley, 2000.
- [16] M. Magdon-Ismail, "NP-hardness and inapproximability of sparse PCA," *Information Processing Letters*, vol. 126, pp. 35–38, 2017.
- [17] P. W. Sauer and M. A. Pai, *Power System Dynamics and Stability*. Upper Saddle River, NJ: Prentice Hall, 1998.
- [18] R. M. Karp, "Reducibility among combinatorial problems," in *Complexity of Computer Computations*, R. E. Miller and J. W. Thatcher, Eds. New York: Plenum, 1972, pp. 85–103.
- [19] S. Poljak and J. Rohn, "Checking robust nonsingularity is NP-hard," *Mathematics of Control, Signals, and Systems*, vol. 6, no. 1, pp. 1–9, 1993.
- [20] A. Nemirovskii, "Several NP-hard problems arising in robust stability analysis," *Mathematics of Control, Signals, and Systems*, vol. 6, no. 2, pp. 99–105, 1993.
- [21] V. Katewa and F. Pasqualetti, "On the real stability radius of sparse systems," *Automatica*, vol. 113, p. 108685, 2020.
- [22] A. Brauer, "Limits for the characteristic roots of a matrix. IV: Applications to stochastic matrices," *Duke Mathematical Journal*, vol. 19, no. 1, pp. 75–91, 1952.
- [23] T. Athay, R. Podmore, and S. Virmani, "A practical method for the direct analysis of transient stability," *IEEE Transactions on Power Apparatus and Systems*, vol. PAS-98, no. 2, pp. 573–584, 1979.
- [24] H. Cui, F. Li, and K. Tomsovic, "Hybrid symbolic-numeric framework for power system modeling and analysis," *IEEE Transactions on Power Systems*, vol. 36, no. 2, pp. 1373–1384, 2021.
- [25] ANDES Development Team, "ANDES IEEE 39-bus test case," https://docs.andes.app/en/v1.9.3/getting_started/testcases/index.html, version 1.9.3 test-case index; dynamic case introduced in v1.8.7; accessed Aug. 28, 2026.
- [26] Y. A. Kuznetsov, *Elements of Applied Bifurcation Theory*, 3rd ed. New York: Springer, 2004.
- [27] J. J. Sanchez-Gasca, V. Vittal, M. J. Gibbard, A. R. Messina, D. J. Vowles, S. Liu, and U. D. Annakkage, "Inclusion of higher order terms for small-signal (modal) analysis: Committee report – task force on assessing the need to include higher order terms for small-signal (modal) analysis," *IEEE Transactions on Power Systems*, vol. 20, no. 4, pp. 1886–1904, 2005.
- [28] I. J. Pérez-Arriaga, G. C. Verghese, and F. C. Schweppe, "Selective modal analysis with applications to electric power systems, part I: Heuristic introduction," *IEEE Transactions on Power Apparatus and Systems*, vol. PAS-101, no. 9, pp. 3117–3125, 1982.
- [29] C. Cañizares *et al.*, "Benchmark systems for small-signal stability analysis and control," IEEE Power and Energy Society, Technical Report PES-TR18, 2015.